13.4 Mean or Median?

WORKSHEET

The question

The random variables $X_1, X_2, \ldots, X_{2n+1}$ are independently and uniformly distributed on the interval $0 \leq x \leq 1$. The random variable Y is defined to be the median of $X_1, X_2, \ldots, X_{2n+1}$. Given that the probability density function of Y is $g(y)$, where

$$g(y) = \begin{cases} ky^n(1-y)^n & \text{if } 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

use the result

$$\int_0^1 y^r(1-y)^s \, dy = \frac{r!\, s!}{(r+s+1)!}$$

to show that $k = (2n+1)!/(n!)^2$, and evaluate $E(Y)$ and $\mathrm{Var}(Y)$. Hence show that, for any given positive number d, the inequality

$$P\left(|Y - \tfrac{1}{2}| < \frac{d}{\sqrt{n}}\right) < P\left(|\bar{X} - \tfrac{1}{2}| < \frac{d}{\sqrt{n}}\right)$$

holds provided n is large enough, where $\bar{X}$ is the mean of $X_1, X_2, \ldots, X_{2n+1}$.

[You may assume that Y and $\bar{X}$ are normally distributed for large n.]

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Why is the density of the median proportional to $y^n(1-y)^n$? (For Y to be near y, how many of the $2n+1$ values must be below y, and how many above?)

STAGE 2 Mean and variance

Ask yourself Compute $\mathrm{E}(Y^2) = k\int_0^1 y^{n+2}(1-y)^n\,\mathrm{d}y$ using the beta integral, and simplify the factorials.

STAGE 3 Comparing with the mean

Ask yourself What are the mean and variance of $\bar{X}$, the average of $2n+1$ independent uniform variables?

Ask yourself If Y and $\bar{X}$ are normal with mean $\frac{1}{2}$ and standard deviations σ_Y and $\sigma_{\bar{X}}$, write both probabilities in terms of the standard normal distribution function Φ.

Try this next

1. **The median's density.** Derive the density $\frac{(2n+1)!}{(n!)^2}y^n(1-y)^n$ by counting: one value in $[y, y+\delta y]$, n below and n above.
2. **Three observations.** For $n=1$ (three uniform observations), find $\mathrm{Var}(Y)$ and $\mathrm{Var}(\bar{X})$ exactly. Which is smaller?
3. **The midrange.** Another estimator of the centre is the midrange $\frac{1}{2}(\max+\min)$. Show that for uniform data its variance is $\frac{1}{2(m+1)(m+2)}$ with $m=2n+1$ observations. How does it compare with the mean? (Compare Section 13.3.)
4. **An outlier.** Eleven measurements are 9.8, 9.9, 10.0, 10.0, 10.1, 10.1, 10.2, 10.2, 10.3, 10.4 and 104 (a typing error). Compare the mean and median.

Further reading

- **John Rice, *Mathematical Statistics and Data Analysis*** (Duxbury), on order statistics and efficiency.

- **Rand Wilcox, *Fundamentals of Modern Statistical Methods*** (Springer). Why robust methods matter, for non-specialists.
- **Stephen Stigler, *The Seven Pillars of Statistical Wisdom*** (Harvard University Press). Aggregation, the mean and the median, in historical perspective.

The full guided solution, and what the question leads to at university, is in Section 13.4:
https://beyondstep.keithmcnulty.org/ch13/mean-or-median.html